Modified DoubleUNet: An Improved Medical Image Segmentation Deep Learning Model

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

by

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled **"Modified DoubleUNet: An Improved Medical Image Segmentation Deep Learning Model"** submitted by **Aditya Apoorva Negi** bearing the **MIS No: 112315010**, **Aditya Harikrishnan** bearing the **MIS No: 112315011**, **M. Arsalan Siddiqui** bearing the **MIS No: 112315108**, **Jonthan Jesni** bearing the **MIS No: 112315078,** and **Om Arun Thombre** bearing the **MIS No: 112315119** in completion of their project work under the guidance of **Dr. Mohammad Aklaqur Rahman** is accepted for the project report submission in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in the **Department of Computer Science and Engineering**, Indian Institute of Information Technology, Pune (IIIT Pune), during the academic year **2025-26**.

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Abstract

Semantic segmentation is a cornerstone of computer-aided diagnostic systems, enabling pixel-level classification essential for precise biomedical image analysis. While traditional encoder-decoder architectures like U-Net have set the standard for capturing contextual and spatial details, enhancing performance on complex medical structures remains a priority.

In this work, we propose and adapt a Modified Double U-Net architecture designed for multi-class breast cancer segmentation using the BUSI (ultrasound) and CBIS-DDSM (mammography) datasets. Building upon the sequential stacking of two U-Net frameworks, our model leverages pretrained encoders (VGG-19, XCeption, and DenseNet) to extract rich hierarchical features, which are subsequently refined by a second U-Net stage.

The Modified Double U-Net was evaluated for binary segmentation tasks, demonstrating the significant impact of stacked architectural refinement on medical image analysis. The model achieved a superior Mean Intersection over Union (IoU) of 95.94% and a Mean Dice score of 97.93%, proving highly effective at general tumor localization. This architectural superiority is driven by the integration of a pretrained ensemble encoder (VGG-19, DenseNet, and Xception), which enhances feature extraction and allows the model to capture complex patterns even with limited data. By utilizing a two-stage refinement process, the second U-Net acts as a specialized refiner for the initial prediction mask, leading to superior boundary localization compared to single-stage baselines like U-Net and Double U-Net. Unlike the original binary Double U-Net, we extend the architecture to perform ternary segmentation, distinguishing between background, benign lesions, and malignant lesions to provide more clinically nuanced interpretations. The Modified Double U-Net was evaluated for ternary segmentation on breast imaging datasets, demonstrating the significant impact of stacked architectural refinement on multi-class medical image analysis. The model achieved competitive performance across modalities, reaching a Mean IoU of 77.77% and a Mean Dice score of 82.84% on the BUSI dataset, while maintaining an IoU of 59.23% and a Dice score of 73.37% on the more complex CBIS-DDSM mammography dataset. The model maintains a stable precision-recall balance, yet the variance in average metrics across datasets suggests that the high performance is partially influenced by the accurate delineation of dominant background structures. These findings indicate that while the stacked architecture provides a powerful foundation for lesion localization, future refinements should focus on advanced class-balancing strategies, such as weighted loss functions or targeted oversampling, to further improve the precision of multi-class boundary delineation across diverse medical imaging modalities.

Keywords: Convolutional Neural Network, Medical Image Segmentation, Double U-Net, Multi-class Segmentation, Tumor Segmentation, Transfer Learning.

TABLE OF CONTENTS

List of Nomenclature

Semantic Segmentation
Pixel-wise classification where each pixel is assigned a class label.

Multi-class Segmentation
Segmentation with more than two classes (e.g., background, benign, malignant).

CNN (Convolutional Neural Network)
A deep learning architecture designed for image processing that uses convolution operations to automatically learn spatial features such as edges, textures, and patterns. CNNs consist of layers like convolution, activation, pooling, and fully convolutional layers, enabling hierarchical feature extraction from low-level (edges) to high-level (objects).

Activation Function (ReLU — Rectified Linear Unit)
A non-linear function applied after convolution to introduce non-linearity into the model, allowing it to learn complex patterns. ReLU is computationally efficient and helps avoid vanishing gradient problems, making it widely used in deep networks.

Decoder (Expanding Path)
The part of the network responsible for reconstructing the segmentation map by increasing spatial resolution. It uses upsampling and combines features from the encoder to recover fine-grained spatial details and produce pixel-level predictions.

Max Pooling
A downsampling operation that reduces the spatial dimensions of feature maps by selecting the maximum value from a small region (e.g., 2×2 window). It helps reduce computation, control overfitting, and retain the most important features while discarding less relevant information.

Encoder (Contracting Path)
The part of the network that progressively reduces spatial dimensions while increasing feature depth to extract contextual and semantic information from the image. It typically consists of repeated convolution + ReLU + pooling operations and acts as a feature extractor.

Upsampling / Up-Convolution
A process used in the decoder to increase the spatial resolution of feature maps. In this work, upsampling is performed using bilinear interpolation followed by convolutional layers, allowing stable and smooth reconstruction of spatial features while avoiding artifacts associated with transposed convolutions.

Bilinear Interpolation
A non-learnable upsampling technique used to increase the spatial resolution of feature maps by estimating new pixel values based on a weighted average of the four nearest neighboring pixels. It performs interpolation in both horizontal and vertical directions, resulting in smoother and more continuous outputs compared to nearest-neighbor interpolation.

Skip Connections
Direct connections between corresponding encoder and decoder layers that transfer high-resolution spatial information lost during downsampling. They help improve localization

accuracy and prevent information loss, which is critical for segmentation tasks.

1×1 Convolution
A convolution operation with a 1×1 kernel used to reduce or map feature channels to the desired number of output classes. It acts as a channel-wise transformation and is commonly used in the final layer to generate segmentation masks.

Kernel/Filter
Small matrix used to extract patterns from images

Transfer Learning
A technique where a model uses weights pretrained on a large dataset (e.g., ImageNet) and applies them to a new task. It improves performance, reduces training time, and helps when the dataset is small by reusing learned features like edges and shapes.

VGG-19
A deep CNN architecture with 19 layers that uses small (3×3) convolution filters stacked deeply. It is simple, uniform, and effective for feature extraction, making it widely used in transfer learning for segmentation models.

DenseNet — Densely Connected Convolutional Network
A convolutional neural network architecture in which each layer is directly connected to all preceding layers within a dense block. DenseNet uses feature concatenation, allowing each layer to access the collective knowledge of all previous feature maps. DenseNet is particularly effective as an encoder backbone in segmentation tasks, as it preserves fine-grained features and improves representation learning even with limited medical data.

Xception — Extreme Inception Network
A deep convolutional neural network architecture that replaces standard convolutions with depthwise separable convolutions, decoupling spatial filtering and channel-wise feature learning. Instead of performing convolution jointly across spatial and channel dimensions, Xception splits it into two steps:

1. Depthwise convolution → applies spatial filtering independently on each channel
2. Pointwise (1×1) convolution → combines channel information

This results in reduced computational cost (fewer parameters and operations), better feature extraction efficiency, and improved performance on complex visual tasks.

Ensemble Learning
A technique that combines multiple models or feature extractors to improve overall performance.

Squeeze-and-Excitation (SE) Block
A mechanism that adaptively recalibrates channel-wise feature responses by emphasizing important features and suppressing less useful ones. It improves model performance by focusing attention on the most relevant information in feature maps.

ASPP — Atrous Spatial Pyramid Pooling
A module that uses multiple parallel dilated (atrous) convolutions with different dilation rates to capture multi-scale context. It allows the network to understand both small and large objects without increasing computational cost significantly.

List of Formulae

Convolution Operation

$$y(i,j) = \sum_m \sum_n x(i+m, j+n) \cdot w(m,n)$$

ReLU Activation

$$ReLU(x) = \max(0, x)$$

Max Pooling

$$y = \max(x_{region})$$

DenseNet Connectivity

$$x_l = H_l([x_0, x_1, x_2, ..., x_{l-1}])$$

Standard Convolution

$$Cost = D_k^2 \cdot M \cdot N \cdot D_f^2$$

Depthwise Separable Convolution

$$Cost = D_k^2 \cdot M \cdot D_f^2 + M \cdot N \cdot D_f^2$$

Bilinear Interpolation

$$f(x,y) = \sum_{i=0}^{1} \sum_{j=0}^{1} w_{ij} \cdot Q_{ij}$$

Dilated Convolution

$$y[i] = \sum_k x[i + r \cdot k] \cdot w[k]$$

Squeeze (Global Average Pooling)

$$z_c = \frac{1}{H \cdot W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_c(i,j)$$

Excitation

$$s = \sigma(W_2 \cdot ReLU(W_1 \cdot z))$$

Scaling

$$\tilde{x}_c = s_c \cdot x_c$$

Binary Cross Entropy Loss

$$L = -[y \log(p) + (1-y) \log(1-p)]$$

Multi-class Cross Entropy Loss

$$L = -\sum_{i=1}^{C} y_i \log(p_i)$$

Softmax Function

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}}$$

IoU (Jaccard Index)

$$IoU = \frac{TP}{TP+FP+FN}$$

Dice (F1 Score)

$$Dice = \frac{2TP}{2TP + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall

$$Recall = \frac{TP}{TP + FN}$$

List of Figures

List of Tables

Chapter 1
Introduction

1.1 Overview of Work

The advancement of computer-aided diagnosis (CAD) systems has significantly improved the ability of clinicians to detect and analyze diseases using medical imaging. Automated image processing techniques reduce manual effort, minimize diagnostic errors, and enable faster clinical decision-making. Among these techniques, image segmentation plays a critical role by assigning a class label to each pixel in an image, allowing precise localization of pathological regions such as tumors or lesions.

Over the years, various traditional segmentation methods have been proposed, including threshold-based techniques [1], [2], region growing [3], [4], machine learning approaches [5], [7], and active contour models [8], [9]. While these methods provide useful clinical support, they often struggle with complex image characteristics such as noise, low contrast, and variability in medical datasets.

The emergence of deep learning, particularly convolutional neural networks (CNNs), has significantly transformed visual recognition tasks [6]. Unlike traditional classification tasks that produce a single output per image, medical image segmentation requires pixel-wise predictions. Fully Convolutional Networks (FCNs) were among the first deep learning architectures designed for this purpose [7], followed by U-Net, which introduced an encoder–decoder structure with skip connections to effectively capture both spatial and contextual information [8].

Further improvements were introduced with Double U-Net, which stacks two U-Net architectures sequentially to refine segmentation outputs and improve performance [9]. Building upon this foundation, this project implements a modified Double U-Net architecture for breast image segmentation using the BUSI (ultrasound) and CBIS-DDSM (mammography) datasets.

Unlike the original architecture, which is primarily designed for binary segmentation, this work extends the model to perform multi-class segmentation, distinguishing between background, benign lesions, and malignant lesions. Additionally, the model incorporates ensemble learning by combining pretrained encoders such as VGG-19, DenseNet, and Xception, leveraging transfer learning from large-scale datasets like ImageNet [11] to improve feature extraction and generalization.

1.2 Literature Review

A substantial body of research exists in the field of medical image segmentation, particularly with the evolution from traditional image processing techniques to deep learning-based architectures. Table 1.1 summarizes key studies and their contributions relevant to this project.

Table 1.1: Summary of Key Literature Reviewed in Medical Image Segmentation

Study	Method Used	Key Findings / Takeaways
Ilhan & Ilhan (2017) [1]; Tamal (2020) [2]	Threshold-based segmentation	Simple and computationally efficient methods, but highly sensitive to intensity variations and noise, limiting performance in complex medical images.
Rundo et al. (2016) [3]; Biratu et al. (2021) [4]	Region Growing	Effective for homogeneous regions but dependent on seed selection and prone to leakage in noisy images.
Vadhnani & Singh (2022) [5]; Militello et al. (2022) [7]	Machine Learning-based segmentation	Improved performance over classical methods but relies heavily on handcrafted features and lacks robustness across datasets.
Dake et al. (2019) [8]; Shahvaran et al. (2021) [9]	Active Contour Models	Capable of capturing object boundaries accurately, but computationally expensive and sensitive to initialization.
Krizhevsky et al. (2012) [6]	Deep CNNs	Demonstrated the effectiveness of deep learning in visual recognition, enabling automatic feature extraction and improved performance.
Long et al. (2015) [7]	Fully Convolutional Networks (FCN)	First deep learning architecture for pixel-wise segmentation; enabled end-to-end learning but lacked precise localization.
Ronneberger et al. (2015) [8]	U-Net (Encoder–Decoder with Skip Connections)	Significantly improved biomedical segmentation by combining contextual and spatial features; became a standard architecture.

Jha et al. (2020) [9]	Double U-Net	Introduced a two-stage segmentation approach that refines outputs, improving segmentation accuracy over U-Net.
Deb et al. (2022) [16]; Deb et al. (2020) [15]	Ensemble Learning	Combining multiple models improves performance, robustness, and generalization compared to single-model approaches.
Deng et al. (2009) [11]; Simonyan et al. (2014) [10]; He et al. (2016) [17]	ImageNet Pretraining (Transfer Learning)	Pretrained models improve feature extraction and reduce training requirements for smaller datasets.

The literature indicates a clear progression from traditional segmentation techniques to deep learning-based architectures, with recent works emphasizing encoder–decoder models and ensemble learning. However, limitations such as binary segmentation focus and lack of multi-class adaptation motivate the proposed work.

1.3 Motivation of the Work

The rapid advancement of medical imaging technologies has significantly increased the availability of breast imaging data, particularly in modalities such as ultrasound (BUSI) and mammography (CBIS-DDSM). While this provides valuable opportunities for early disease detection, it also introduces challenges related to manual analysis, diagnostic consistency, and time efficiency. These limitations highlight the need for automated and reliable segmentation systems.

The following factors motivate the development of this project:

- **Overcoming the Limitations of Early Segmentation Techniques**
  Manual segmentation of medical images is time-consuming and highly subject to inter-observer variability. Early automated methods, such as thresholding, active contours, and region growing, relied heavily on raw pixel intensities and mathematical rules [1]–[4]. These methods frequently fail in breast imaging due to low contrast and overlapping tissue densities. While baseline deep learning architectures like U-Net revolutionized this process by using encoder-decoder structures and skip connections to recover spatial details [8], modern clinical demands require even higher precision.

- **Addressing "Data Starvation" and Complex Boundaries**
  Medical datasets are notoriously small compared to standard computer vision datasets. Training a standard U-Net from scratch on limited breast imaging data often leads to overfitting. Furthermore, U-Net performs only a single pass to reconstruct an image, which is often insufficient for resolving the fuzzy, irregular boundaries of breast lesions. This motivated the shift toward advanced stacked architectures like Double U-Net, which introduced transfer learning (using a pre-trained VGG-19 encoder) to solve data starvation, and iterative refinement (stacking two U-Nets) to progressively smooth and correct tumor boundaries [9].

- **Breaking the "Single-Model" Bottleneck via Ensemble Learning**
  While Double U-Net improved upon baseline models, it introduced a new bottleneck by relying exclusively on a single pre-trained network (VGG-19) for initial feature extraction. A single backbone may fail to capture the highly diverse and heterogeneous textures found across different imaging modalities like BUSI and CBIS-DDSM. This project is motivated by the need for superior feature representation, achieved through the Modified Double U-Net architecture. By replacing the single encoder with an ensemble of pre-trained networks (Xception, DenseNet121, and VGG-19) and upgrading the skip connections to pull high-resolution data from multiple encoders simultaneously [10], [13], [14], the model becomes significantly more robust to noise and varied anatomical structures.

- **The Need for Multi-Class Clinical Interpretation**
  A critical limitation of existing segmentation systems, including the original U-Net, standard Double U-Net, etc., is their inherent restriction to binary classification (lesion vs. background) [8], [9]. In real-world oncological scenarios, merely localizing a mass is insufficient; distinguishing between benign fibroadenomas and malignant carcinomas is crucial for immediate treatment planning. This project is heavily motivated by the need to bridge this clinical gap by extending the Modified Double U-Net to perform ternary segmentation. By adapting the final output layers to simultaneously classify and delineate background, benign, and malignant tissues, the proposed system provides a significantly higher level of diagnostic utility and clinical interpretability.

Together, these motivations justify the development of the Modified Double U-Net as a robust and efficient deep learning-based segmentation framework, designed to accurately identify and differentiate breast lesion regions in medical images while ensuring improved feature representation, generalization, and clinical relevance.

How our Modified DoubleUNet Addresses These Motivations

- Extending Double U-Net to multi-class segmentation (background, benign, malignant)
- Integrating ensemble learning using VGG-19 [10], DenseNet [14], and Xception [13]
- Applying transfer learning using ImageNet-pretrained weights [11]
- Combining BUSI[19] and CBIS-DDSM[18] datasets into a unified pipeline
- Focusing on practical implementation and reproducibility

By integrating ensemble-based feature extraction, multi-scale contextual learning, and multi-class segmentation into a unified framework, the proposed Modified Double U-Net effectively addresses key limitations in existing segmentation approaches and enhances performance on real-world breast imaging datasets.

Chapter 2
Problem Statement

Existing medical image segmentation systems rely predominantly on conventional deep learning architectures such as U-Net and its variants for tumor localization. While these models have shown promising results, they often struggle to provide accurate, robust, and clinically reliable segmentation in complex medical imaging scenarios.

Insights from prior research indicate that traditional image processing techniques lack generalization and require manual intervention [1]–[5], while early deep learning models such as Fully Convolutional Networks (FCNs) suffer from loss of spatial information [7]. U-Net improves localization through encoder–decoder architecture and skip connections [8], but still faces challenges in handling low-contrast images, noise, and irregular tumor boundaries.

Advanced models like Double U-Net [9] attempted to enhance segmentation by stacking multiple networks and incorporating a pretrained encoder. However, the original Double U-Net architecture suffers from several critical structural limitations. First, it relies on a single pretrained network (VGG-19) for its initial feature extraction, which limits the model to the specific mathematical patterns that single architecture is best at detecting. Second, as images are compressed and decompressed through the stacked networks, the model often loses exact spatial details, leading to less precise boundaries. Finally, the original architecture is inherently limited to binary segmentation, significantly reducing its clinical interpretability.

These limitations contribute to several persistent challenges in medical image segmentation:

- **Inconsistent Segmentation Accuracy**, especially in small, irregular, or low-contrast tumor regions where spatial details are lost.
- **Poor Generalization Across Datasets**, due to the "single-model" bottleneck struggling to adapt to the high variability in medical imaging modalities.
- **Limited Multi-Class Segmentation Capability**, reducing clinical interpretability by only separating background from foreground.
- **Sensitivity to Noise and Class Imbalance**, particularly in ultrasound images [19].

The core problem this project addresses is: To design and implement an efficient Modified Double U-Net-based segmentation system that overcomes the limitations of existing U-Net and Double U-Net models. The proposed system will improve initial feature extraction by utilizing an ensemble of pretrained encoders (Xception, DenseNet121, and VGG-19) to capture a more diverse set of features. Furthermore, it will enhance spatial precision by upgrading the skip connections in the second decoder to pull high-resolution data from both encoders simultaneously. Ultimately, this enhanced architecture will be adapted to enable accurate multi-class tumor segmentation (e.g., distinguishing between background, benign, and malignant lesions) for complex breast imaging datasets such as BUSI[19] and CBIS-DDSM[18].

2.1 Research Objectives

The primary objective of this project is to design and develop an advanced deep learning-based medical image segmentation system capable of generating accurate, robust, and clinically meaningful tumor segmentation for breast imaging datasets. The system aims to improve segmentation performance across heterogeneous datasets such as CBIS-DDSM [18] and BUSI [19] using an enhanced Modified Double U-Net architecture. The core research goals addressed in this work are outlined below:

1. **Development of a Multi-Class Segmentation Framework**
Design a segmentation system capable of performing pixel-wise classification into multiple clinically relevant categories: background, benign tumor, and malignant tumor.
Extend traditional binary segmentation approaches to multi-class segmentation to improve diagnostic interpretability and clinical usefulness.

2. Enhanced Feature Extraction using Modified Double U-Net
Implement a Modified Double U-Net architecture by leveraging stacked U-Net structures with improved encoder design. Incorporate transfer learning using pretrained networks such as VGG-19 [10], DenseNet [14], and Xception [13] to improve feature representation and generalization. Utilize architectural enhancements such as skip connections, encoder–decoder pathways [8], and multi-scale context extraction to improve segmentation accuracy, especially in low-contrast and noisy images.

3. **Optimization for Accurate and Efficient Segmentation**
Design training strategies to improve convergence, reduce overfitting, and enhance model stability. Optimize loss functions and evaluation metrics such as Dice Coefficient and Intersection over Union (IoU) to ensure high segmentation accuracy. Balance performance improvement with computational efficiency to make the model feasible for practical applications.

4. System Integration, Implementation, and Evaluation
Integrate all components—including dataset preparation, preprocessing, model architecture, training pipeline, and prediction modules—into a unified implementation framework. Evaluate the system using standard performance metrics such as Dice Score, IoU, Precision, and Recall to assess segmentation quality [9].
Analyze model performance across different datasets and lesion types to validate improvements over baseline models such as U-Net [8] and Double U-Net [9].

2.2 Analysis and Design

A detailed analysis of existing medical image segmentation techniques, deep learning architectures, and the limitations identified in the research gap guided the design of the proposed system. The system is structured as a deep learning-based segmentation pipeline that leverages the strengths of U-Net [8], and Double U-Net [9] architectures, while addressing challenges such as noisy medical images, class imbalance, and variability across imaging modalities.

2.2.1 System Requirement Analysis

The system must achieve the following requirements to be considered a valid solution:

1. **Accurate Multi-Class Segmentation**
   Perform pixel-wise classification into three classes:
   1. Background (0)
   2. Benign lesion (1)
   3. Malignant lesion (2)

   Unlike traditional binary segmentation, the system must distinguish between benign and malignant tumor regions to improve clinical relevance.

2. **Robust Feature Extraction**
   Extract meaningful spatial and contextual features from medical images using deep convolutional architectures. Leverage pretrained encoders such as VGG-19 [10], DenseNet [14], and Xception [13] to improve feature learning and generalization across datasets.

3. **Handling Noisy and Low-Contrast Images**
   Effectively segment images with poor contrast and high noise levels, especially in ultrasound datasets such as BUSI [19].
   Ensure robustness to irregular tumor boundaries and varying image quality.

4. **Dataset Generalization Capability**
   Support training and evaluation across heterogeneous datasets:
   - Ultrasound images (BUSI [19])
   - Mammography images (CBIS-DDSM [18])

   Ensure that the model maintains consistent performance across different imaging modalities.

5. **Efficient Training and Convergence**
   Implement a training pipeline that ensures:
   - Stable convergence
   - Reduced overfitting
   - Efficient use of computational resources

   Use appropriate loss functions and optimization strategies to improve segmentation accuracy.

6. **Mask Generation and Visualization**
   Generate segmentation masks for each input image and provide:
   - Binary masks for training
   - Color-coded masks (background, benign, malignant) for visualization

   Support overlay of predicted masks on original images for qualitative evaluation.

2.2.2 Architectural Design Principles

To satisfy these requirements, the system architecture was designed around the following principles:

1. **Encoder–Decoder Architecture**
   Adopt the U-Net-based encoder–decoder structure [8] to capture both contextual and spatial information. Encoders extract deep semantic features, while decoders reconstruct high-resolution segmentation maps.

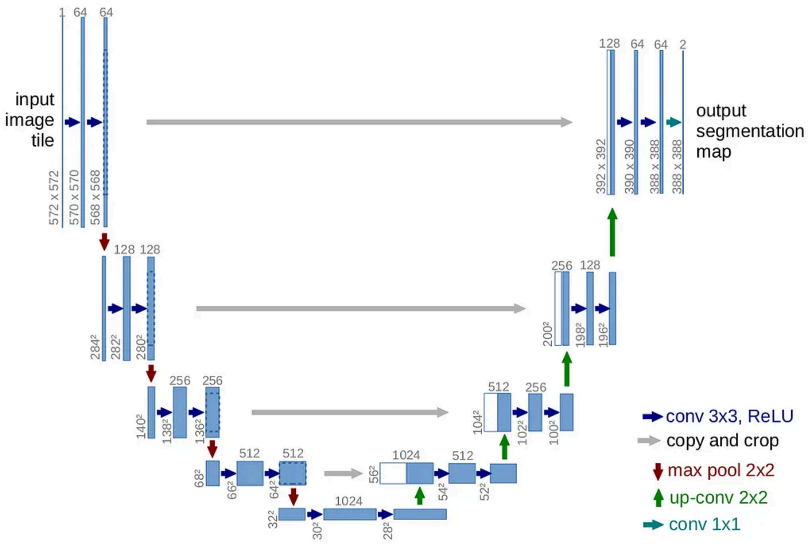

Fig 2.1: U-Net Encoder-Decoder Architecture

2. **Multi-Stage Segmentation Refinement**
   Utilize the Double U-Net concept [9], where:
   - The first network generates an initial segmentation mask
   - The second network refines the segmentation using the original image and intermediate output

   This improves segmentation accuracy, especially for complex and small lesions.

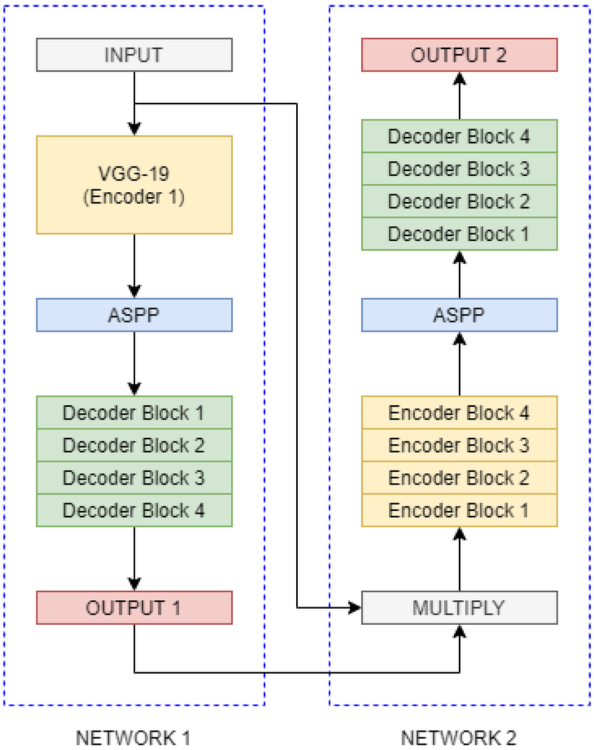

Fig 2.2: Double U-Net

3. **Transfer Learning for Feature Enhancement**
   Incorporate pretrained models such as:
   - VGG-19 [10]
   - DenseNet [14]
   - Xception [13]

   to enhance feature extraction and improve generalization on limited medical datasets.

4. **Multi-Scale Context Capture**
   Integrate mechanisms such as ASPP (Atrous Spatial Pyramid Pooling) to capture features at multiple scales, improving detection of tumors of varying sizes [12].

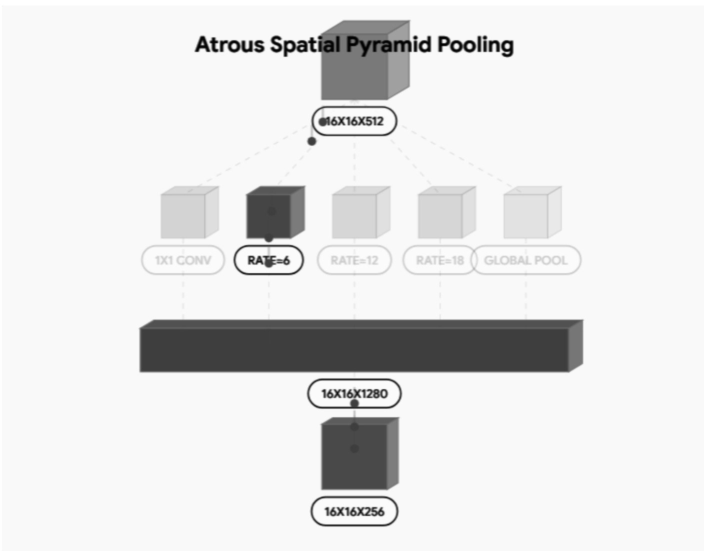

Fig 2.3: ASPP Module Flow

5. **Feature Fusion and Skip Connections**
   Use skip connections between encoder and decoder layers to:
   - Preserve spatial information
   - Improve localization accuracy
   - Reduce information loss

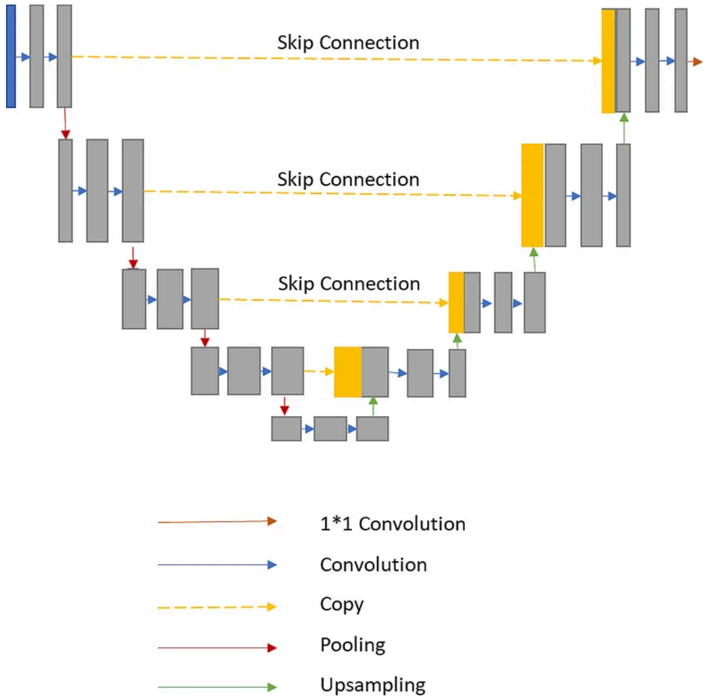

Fig 2.4: Skip Connections, Copy-Crop-Concat Flow

6. **Multi-Class Output Design**
   Design the final layer to output multi-class segmentation masks instead of binary output. Ensure compatibility with class labels:
   - 0 → Background
   - 1 → Benign
   - 2 → Malignant
7. **Data-Driven Optimization**
   Incorporate:
   - Data augmentation
   - Normalization
   - Balanced sampling

   to improve training robustness and reduce dataset bias.

2.2.3 Overall System Architecture

The overall architecture integrates the above principles into a cohesive pipeline with the following major components:

1. Dataset Preparation Module
2. Preprocessing and Augmentation Module
3. Model Architecture Module
4. Training Module
5. Prediction and Visualization Module
6. Evaluation Module

The system has been carefully designed to address the shortcomings identified in the research gap, leveraging advanced deep learning techniques and multi-stage feature extraction to improve segmentation performance. By combining robust encoder–decoder architectures, transfer learning, and multi-class segmentation capabilities, the proposed system is aligned with the complexities of heterogeneous medical imaging datasets such as BUSI and CBIS-DDSM. This analysis and architectural design form the foundation for the detailed methodology and implementation described in Chapter 3.

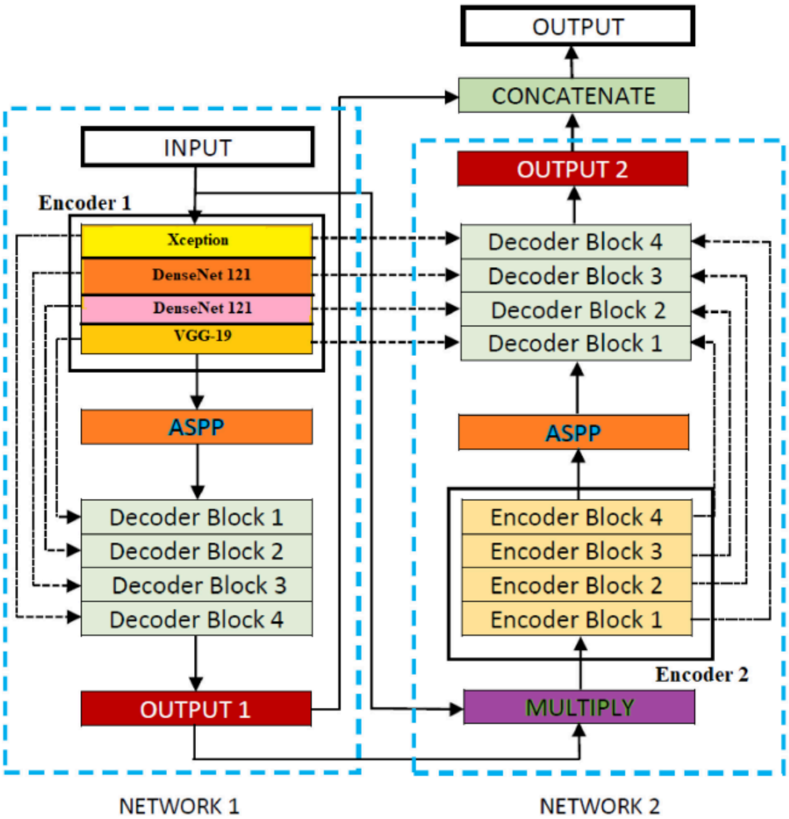

Fig 2.5: Modified System Architecture

Chapter 3
Proposed Work

The proposed work aims to design, implement, and evaluate a deep learning-based medical image segmentation system using a Modified Double U-Net architecture, tailored for breast tumor analysis on ultrasound and mammography datasets. The system is implemented as a practical Python-based pipeline that processes medical images from raw dataset format to final segmentation outputs, including multi-class mask prediction.

This chapter presents the core components of the implemented system, including the methodology of work, modular pipeline design, dataset handling scripts, preprocessing logic, model architecture implementation, training configuration, and prediction workflow. Each module is described in detail based on the actual Python scripts developed in this project—such as dataset preparation, training, and inference—to illustrate how the system addresses the identified research gaps and delivers a robust, accurate, and practically deployable segmentation solution.

3.1 Methodology of Work

3.1.1 Data Acquisition and Dataset Preparation

A dataset preparation script is implemented to convert raw dataset formats into a structured segmentation dataset.
Key steps include:

- Image–Mask Pair Extraction
  Mapping each image to its corresponding segmentation mask.
  - 0 → Background
  - 1 → Benign lesion
  - 2 → Malignant lesion
- Dataset Structuring
  Creation of organized directories:
  - train/
  - val/
  - test/
- Mask Generation
  - Binary masks for training
  - Color masks for visualization (black, green, red)
- Data Cleaning
  - Removal of inconsistent samples
  - Handling missing or duplicate files

  This stage produces a clean dataset (dataset_seg) suitable for training.

3.1.2 Preprocessing and Data Augmentation

Medical images often contain noise, low contrast, and variability. To improve robustness, preprocessing and augmentation techniques are applied.

- Preprocessing steps:
  - Image resizing to uniform dimensions
  - Normalization of pixel values
  - Mask alignment with corresponding images

- Data augmentation techniques:
    - Rotation
    - Flipping
    - Scaling
    - Elastic transformations

These transformations increase dataset diversity and improve generalization of the model.

3.1.3 Model Architecture Implementation

The system implements a Modified Double U-Net architecture based on encoder–decoder design. In the dual UNet, the first network generates an initial segmentation, and the second network refines the segmentation. VGG-19, Xception, and DenseNet used for feature extraction. Skip Connections preserve spatial information during reconstruction. Multi-Scale Feature Learning (ASPP) improves detection of tumors of varying sizes. The final output layer produces multi-class segmentation masks (3 classes).

3.1.4 Training Pipeline

The training process is implemented through a dedicated Python script (train.py).
Key Components:

- Loss Function
    - Categorical Cross-Entropy / Dice Loss
- Optimizer
    - Adam Optimizer
- Evaluation Metrics
    - Dice Coefficient
    - Intersection over Union (IoU)
    - Precision and Recall
- Training Strategy
    - Train–Validation split
    - Early stopping to prevent overfitting
    - Learning rate scheduling

Table 3.1: Hyperparameters and Configuration of the Modified Double U-Net Architecture

Hyperparameter	**Value**	**Description**
Input Image Size	256 × 256	Size to which input images are resized before feeding into the network
Number of Classes	3	Pixel-wise segmentation classes (multi-class segmentation): Background, Benign, Malignant
Batch Size	4	Number of samples processed before updating weights
Epochs	50-75 with early stopping	Total passes over the training dataset

Optimizer	Adam	Adaptive optimizer for faster convergence
Activation Function	ReLU (hidden layers), Softmax (output)	Introduces non-linearity and class probabilities
Weight Initialization	Pretrained (ImageNet)	Helps improve convergence and generalization
Data Augmentation	Rotation, Flip, Elastic Transform	Increases dataset diversity
Train/Val/Test Split	80 / 10 / 10	
Pooling Type	MaxPooling (2×2)	Downsampling in encoder
Upsampling	Bilinear Conv	Restores spatial resolution in decoder
Kernel Size	3 × 3 (Conv layers)	Standard convolution filter size
Stride	1 (Conv), 2 (Pooling)	Controls movement of filters
Skip Connections	Yes	Skip Connections

3.1.5 Prediction and Inference

The trained model is used to generate segmentation masks for unseen images using the predict.py module.
The output includes:

- Predicted segmentation masks
- Color-coded masks:
    - Background → Black
    - Benign → Green
    - Malignant → Red
- Overlay visualization on original images

This enables qualitative evaluation of model performance.

3.1.6 Evaluation and Performance Analysis

The system performance is evaluated using standard segmentation metrics:

- Dice Coefficient → overlap accuracy
- IoU (Intersection over Union) → segmentation quality
- Precision & Recall → classification reliability

Performance is compared against baseline models such as:

- U-Net [8]
- Double U-Net [9]

3.1.7 System Optimization and Efficiency

To ensure efficient execution, the system incorporates:

- Modular Python scripts (prepare, train, predict)

- Dataset reuse and structured storage
- GPU-based training for faster convergence.

This ensures scalability and practical usability.

3.1.8 Complete Pipeline

The final system follows the sequence::

1. Dataset Acquisition →
2. Dataset Preparation →
3. Preprocessing & Augmentation →
4. Model Training →
5. Prediction →
6. Evaluation →
7. Visualization

This pipeline ensures accurate, robust, and clinically meaningful segmentation outputs.

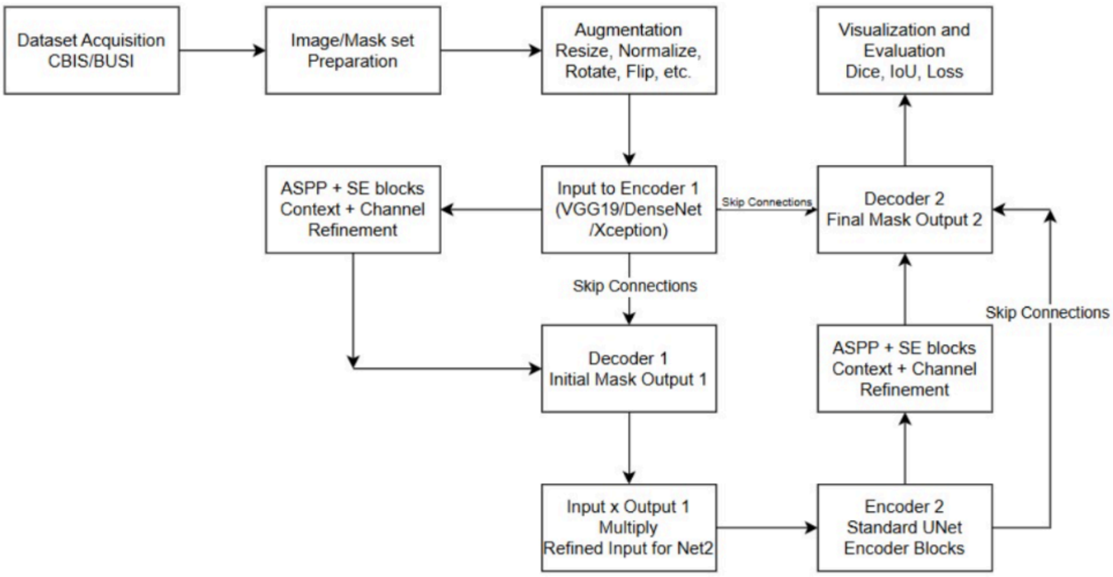

Fig 3.1: Modified DoubleUNet Project Pipeline

3.1.9 System Components and Implementation Modules

The proposed system is implemented using a modular Python-based architecture, where each component is responsible for a specific stage in the segmentation pipeline. This modular design improves maintainability, scalability, and ease of experimentation.

1. Dataset Preparation Module

prepare_busi_dataset.py
prepare_cbis_dataset.py
This module is responsible for converting raw datasets into a structured segmentation-ready format.

2. Model Architecture Module

BUSI_model.py
CBIS_model.py
Defines the Modified Double U-Net architecture used for segmentation. Designed separately for BUSI and CBIS datasets to handle modality-specific variations

3. Training Module
train_BUSI.py
train_CBIS.py
Handles model training and validation. Loads dataset from prepared directories, applies data augmentation and preprocessing steps, and then trains the model using defined loss functions and optimizers. Tracks training and validation performance, and saves trained model weights for inference
4. Prediction and Inference Module
predict_BUSI.py
predict_CBIS.py
Generates segmentation outputs for test images. It loads trained model weights, and produces predicted masks.

5. Evaluation and Testing Module
test_BUSI.py
test_CBIS.py
metrics.py
Evaluates model performance using quantitative metrics. Compares predicted masks with ground truth.

6. Utility and Support Module
utils.py
Provides helper functions used across the system:
- Data loading utilities
- Preprocessing functions
- Image transformations
- Visualization helpers

7. Dataset and Output Storage Structure
BUSI_dataset/ → Raw ultrasound dataset
CBIS_dataset/ → Raw mammography dataset
dataset_seg_BUSI/ → Processed BUSI dataset
dataset_seg/ → Processed CBIS dataset
results_BUSI/ → Prediction outputs (BUSI)
results_CBIS/ → Prediction outputs (CBIS)

8. Configuration and Environment
requirements.txt
.venv/
.gitignore
Defines required libraries and dependencies, maintains an isolated Python environment, ensures reproducibility of experiments.

The proposed system integrates dataset preparation, preprocessing, deep learning-based segmentation, and evaluation into a unified pipeline capable of delivering accurate, robust, and scalable medical image segmentation results. By combining multi-stage feature extraction, encoder–decoder architectures, and multi-class mask prediction, the system effectively captures both spatial and contextual information from medical images. Additionally, the modular implementation and optimized training pipeline ensure efficient processing and adaptability across heterogeneous datasets such as BUSI and CBIS-DDSM. This integrated approach provides a practical and deployable solution that directly addresses the limitations identified in the research gap.

3.2 Dataset Description

Our project constructs a dataset by integrating breast medical imaging data from two primary sources: the BUSI (Breast Ultrasound Images) dataset and the CBIS-DDSM (Curated Breast Imaging Subset of DDSM) dataset. These datasets are further processed using custom Python scripts to generate structured image–mask pairs suitable for deep learning-based segmentation.

Table 3.2: Dataset Overview for Modified DoubleUNet

Dataset Name	Modality	Key Fields	Purpose
BUSI	Ultrasound Images	Breast scans, segmentation masks, class labels (Benign, Malignant, Normal)	Primary training and evaluation for ultrasound-based multi-class segmentation.
CBIS-DDSM	Mammography	Grayscale X-rays, lesion masks, pathology labels (Benign, Malignant)	Cross-modality evaluation and testing generalization on high-resolution X-ray imaging.

1. BUSI Dataset (Breast Ultrasound Images Dataset)

Source: Kaggle [19]
This dataset consists of breast ultrasound images collected for tumor detection and classification. It includes images categorized into benign, malignant, and normal cases, along with corresponding segmentation masks for lesion regions.
Contents
Each sample includes:
- Image — ultrasound scan of the breast
- Mask — binary/annotated tumor region
- Class Label — benign, malignant, or normal

Usage in the System
This dataset is used as the primary source for ultrasound-based segmentation tasks:
- Enables multi-class segmentation:
  - 0 → Background
  - 1 → Benign tumor
  - 2 → Malignant tumor
- Provides real-world examples of:
  - Noisy images
  - Low contrast boundaries
  - Irregular tumor shapes
- Used for:
  - Training the segmentation model
  - Evaluating performance on ultrasound modality
  - Testing robustness against noise and variability

The BUSI dataset plays a critical role in validating the model's ability to handle challenging medical imaging conditions.

2.CBIS-DDSM Dataset (Curated Breast Imaging Subset of DDSM)

Source: Kaggle [18], based on curated dataset from medical imaging research [Nature dataset reference]

This dataset is a curated subset of the Digital Database for Screening Mammography (DDSM), designed to provide standardized and annotated mammography images for breast cancer analysis.

Contents

Each sample includes:

- Mammogram Image — grayscale breast X-ray image
- Segmentation Mask — annotated lesion region
- Pathology Label — benign or malignant
- Lesion Type Information — mass or calcification (if available)

Usage in the System

This dataset is used for mammography-based segmentation and cross-modality evaluation:

- Enables multi-class segmentation similar to BUSI:
    - 0 → Background
    - 1 → Benign lesion
    - 2 → Malignant lesion
- Provides:
    - High-resolution images
    - Structured annotations
    - Clinically validated labels
- Used for:
    - Training on mammography modality
    - Evaluating generalization across datasets
    - Comparing performance between ultrasound and X-ray imaging

The CBIS-DDSM dataset enhances the system's ability to generalize across different imaging techniques and improves clinical applicability.

Chapter 4
Results and Discussion

This chapter presents the evaluation of the segmentation system implemented in this project, based on the Modified Double U-Net architecture. The system is evaluated separately on two datasets:

- BUSI Dataset (Breast Ultrasound Images)
- CBIS-DDSM Dataset (Mammography Images)

The results are analyzed to assess the model's performance in accurately segmenting tumor regions and distinguishing between benign and malignant classes.

Three key evaluation metrics were used:

- Dice Coefficient — measures overlap between predicted and ground truth masks
- Intersection over Union (IoU) — evaluates segmentation accuracy
- Precision and Recall — measure classification reliability at the pixel level

These metrics provide a comprehensive understanding of both segmentation quality and classification performance.

Visual evaluation was also performed using predicted masks and overlay images generated by the system. This allows qualitative assessment of how well the model captures tumor boundaries, especially in challenging cases such as low-contrast ultrasound images and dense mammography scans.

The results presented in this chapter demonstrate the effectiveness of the proposed system in addressing the challenges identified in earlier chapters and highlight its practical applicability in medical image analysis.

4.1 Quantitative Results

To evaluate the performance of the proposed segmentation model, a quantitative analysis was conducted on the prepared breast imaging datasets (BUSI and CBIS-DDSM). The evaluation focuses on pixel-wise segmentation accuracy, where each pixel is classified into one of the defined classes: background, benign lesion, or malignant lesion.

Unlike simple classification tasks, segmentation requires assessing how well the predicted mask overlaps with the ground truth mask. Therefore, standard medical image segmentation metrics were used, including Dice Similarity Coefficient (DSC), Intersection over Union (IoU), Precision, and Recall, which are widely adopted in literature

Each model was evaluated on the test dataset after training, and metric values were computed for every image. The final performance was obtained by averaging these values across all test samples, ensuring a reliable and unbiased evaluation of the model's segmentation capability.

This comprehensive evaluation provides a robust understanding of how effectively the proposed Modified Double U-Net model segments breast lesions under real-world conditions.

Table 4.1: Average Performance of various models for binary segmentation

Model	Parameters	IoU	F1/Dice	Accuracy	Recall
U-Net	+9 million	86.15%	92.56%	94.64%	89.23%
Double U-Net	+24 million	94.36%	97.10%	99.54%	92.86%
Modified Double U-Net	+23 million	95.94%	97.93%	99.65%	92.95%

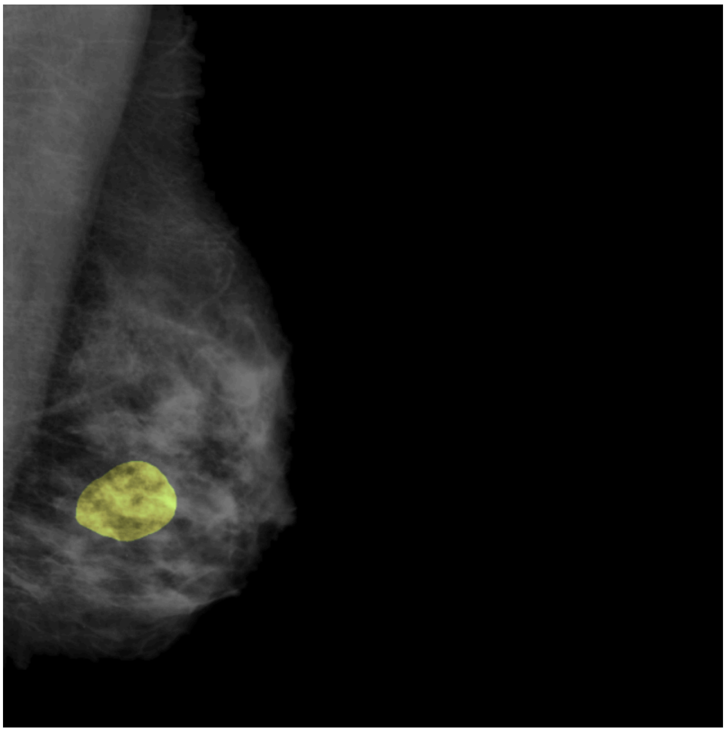

Fig 4.1: Binary Segmentation of Mammogram

Table 4.2: Average Performance of the model for ternary segmentation

Modified Double U Net	Parameters	IoU	F1/Dice	Precision	Recall
CBIS	+49 million	59.23%	73.37%	66.85%	86.60%
BUSI	+49 million	77.77%	82.84%	85.15%	84.64%

Key Findings

- **Superior Architectural Performance**: The Modified Double U-Net achieves the highest overall segmentation performance, demonstrating superior Dice (97.93%) and IoU (95.94%) scores compared to baseline architectures. This confirms that the multi-stage refinement and stacked structure effectively enhance mask quality.
- **Efficacy of Transfer Learning**: The integration of pretrained encoders significantly improves feature extraction, allowing the model to capture complex patterns even with limited medical data. This confirms that transfer learning from ImageNet-trained

models plays a crucial role in improving segmentation accuracy

- **Robustness via Ensemble Learning**: Ensemble-based feature extraction leads to more robust and generalized predictions. By combining multiple CNN architectures, the model learns diverse feature representations, which reduces the risk of overfitting and improves performance across datasets.
- **Two-Stage Refinement**: The Double U-Net structure improves boundary refinement, where the second U-Net enhances the initial mask prediction. This results in better localization of tumor regions compared to single-stage models.
- **Performance vs. Complexity Trade-off**: While the Modified Double U-Net achieves the highest accuracy, it involves a higher number of parameters (49,194,658) in ternary configurations. This increased complexity necessitates higher computational requirements and training time.
- **Cross-Modality Challenges**: Results from multi-class evaluation reveal a performance gap between the BUSI (77.77% IoU) and CBIS-DDSM (59.23% IoU) datasets. While the architecture provides a powerful foundation for lesion localization, the variance across modalities highlights the need for advanced class-balancing strategies, such as weighted loss functions, to master the complex boundaries of malignant classes in high-resolution mammography.

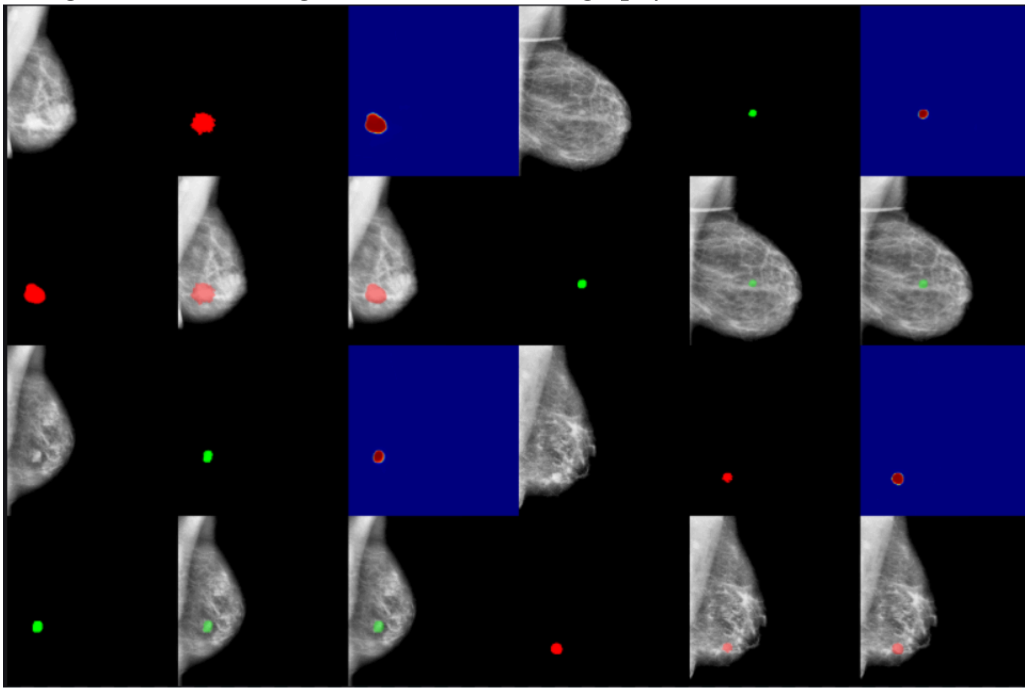

Fig 4.2: CBIS-DDSM Benign vs Malignant Tumor Segmentation

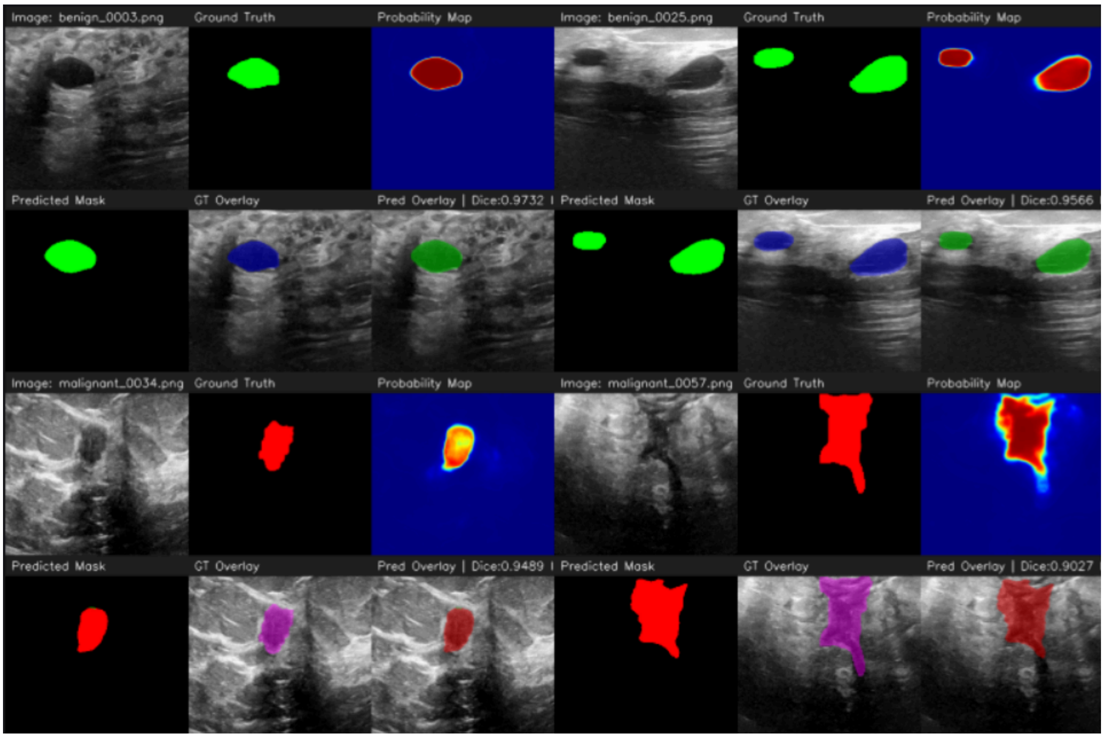

Fig 4.3: BUSI Ultrasound Image Segmentation

4.2 Discussion of Results

Accuracy Metrics – Dice, IoU, Precision, Recall

The model achieves exceptional results in binary segmentation tasks, reaching an IoU of 95.94% and a Dice score of 97.93%. These high values demonstrate a near-perfect degree of overlap between predicted and ground truth masks, validating the model's ability to accurately localize lesion regions. However, a performance gap is observed during the more complex ternary segmentation task. While the model maintains a Mean IoU of 77.77% and a Mean Dice score of 82.84% on the BUSI dataset, metrics shift on the CBIS-DDSM dataset, where the Mean IoU reaches 59.23% and the Mean Dice score is 73.37%. This performance decrease is attributed to the inherent complexity of mammography, which involves:

- Lower contrast levels.
- Higher noise ratios.
- Highly irregular lesion boundaries.

This behavior aligns with Double U-Net research, noting that segmentation performance varies heavily based on dataset complexity and specific image characteristics.

Class-wise Segmentation Performance

A detailed analysis of per-class performance in the ternary task highlights the following insights:

- Background class achieves the highest accuracy (Dice = 0.9771, IoU = 0.9568). This is expected, as background pixels dominate the image and are easier for the model to classify.
- Benign lesions show moderate performance (Dice = 0.5734, IoU = 0.5162). The model detects benign regions reasonably well but struggles with precise boundary definition.
- Malignant lesions demonstrate the lowest performance (Dice = 0.4635, IoU = 0.3866). This is driven by irregular tumor shapes, smaller region sizes, and significant class imbalance.

This trend confirms that multi-class segmentation remains significantly more challenging than binary segmentation in medical imaging.

Precision vs Recall Trade-off

The model maintains a stable balance between precision and recall, particularly in the BUSI ternary evaluation:

- Precision (85.15%): Indicates fewer false positives.
- Recall (84.64%): Suggests the model is effective at identifying lesion regions without significant over-segmentation.

Limitations Observed

Despite competitive performance, several limitations were identified:

- Malignant Class Segmentation: Reduced performance due to complex, infiltrative boundaries.
- Class Imbalance: High mean metrics are partially buoyed by the dominant background class, which achieves near-perfect segmentation (Dice = 0.9771).
- Modality Sensitivity: There is a notable difficulty in handling high-resolution, low-contrast mammography images (CBIS-DDSM) compared to ultrasound scans.
- Computational Complexity: The ensemble architecture utilizes 49,194,658 parameters, which substantially increases training time and computational requirements.

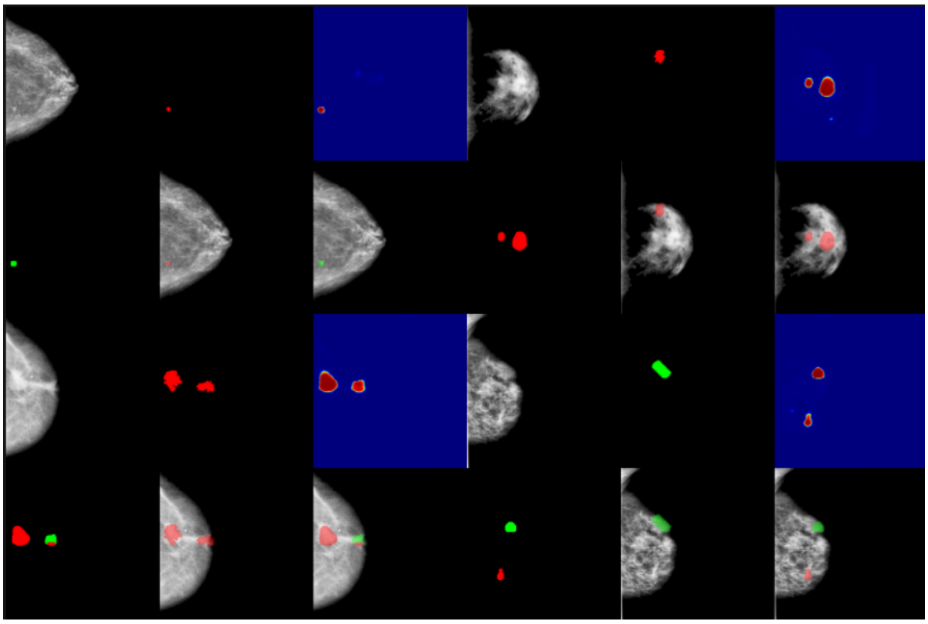

Fig 4.4: Inaccurate Segmentations due to Poor Image Quality or Chaotic Tumor Shapes

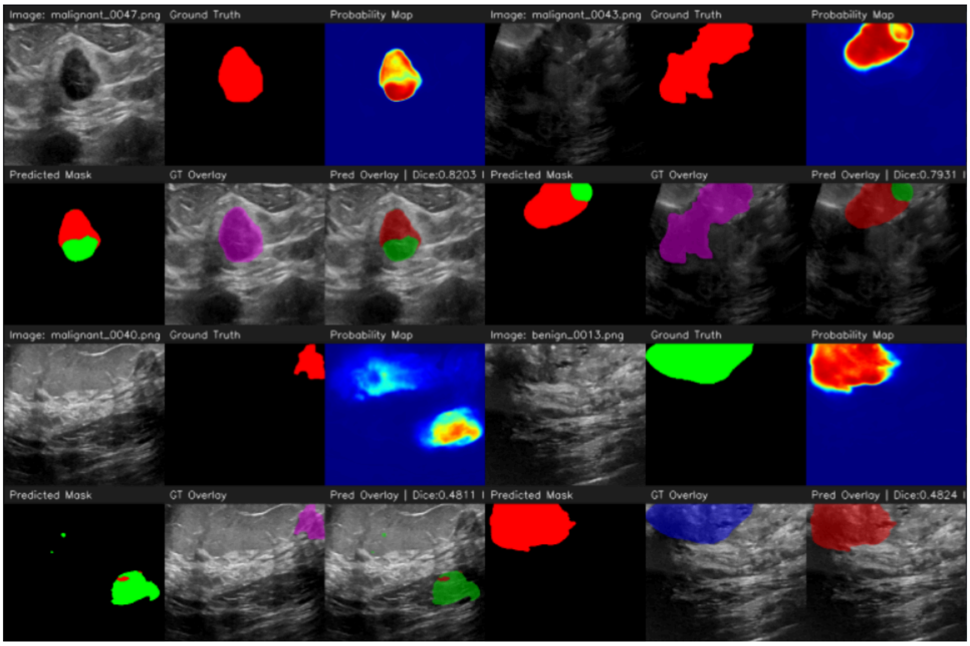

Fig 4.5: Inaccurate Segmentations due to Poor Image Quality or Chaotic Tumor Shapes

4.3 Summary of Findings

Overall Performance of Proposed Model

The Modified Double U-Net based system demonstrates strong and consistent segmentation performance across datasets, with the following key outcomes:

- High Dice and IoU scores for binary segmentation
- Effective two-stage refinement improves mask quality
- Strong feature extraction due to ensemble encoder
- Good balance between precision and recall

Class-wise Insights

- Background class: Highest accuracy
- Benign lesions: Moderate segmentation performance
- Malignant lesions: Most challenging due to complexity and imbalance

Final Conclusion

The proposed Modified Double U-Net–based segmentation system demonstrates clear effectiveness in medical image segmentation tasks, particularly in handling multi-class breast lesion segmentation. While simpler architectures such as U-Net provide reasonable results, the proposed approach significantly improves segmentation quality through multi-stage refinement, ensemble feature extraction, and transfer learning.

The system successfully addresses key challenges identified in the problem statement, including:

- Accurate lesion localization
- Multi-class segmentation capability
- Handling heterogeneous medical imaging data

Although challenges such as class imbalance and dataset complexity remain, the proposed system provides a robust, scalable, and clinically relevant segmentation framework suitable for real-world medical imaging applications.

Chapter 5
Conclusion

Conclusion

This project successfully demonstrates the design and implementation of a deep learning–based medical image segmentation system using a Modified Double U-Net architecture for breast tumor analysis. By leveraging pixel-level learning on medical imaging datasets such as BUSI and CBIS-DDSM, the proposed system effectively segments lesion regions with improved accuracy and robustness compared to traditional approaches.

The model integrates key architectural strengths including dual U-Net stacking, transfer learning through pretrained encoders (VGG-19, DenseNet, etc.), and enhanced feature extraction mechanisms, enabling it to capture both low-level spatial details and high-level semantic information. The use of ensemble-based encoding further strengthens feature representation and improves segmentation performance, particularly in challenging cases involving noisy ultrasound images and complex tumor boundaries .

Additionally, the implementation incorporates multi-class segmentation (background, benign, malignant), making the system more clinically relevant than binary segmentation approaches. This allows not only accurate localization of lesions but also differentiation between tumor types, which is crucial for diagnosis and treatment planning.

Experimental results demonstrate that the proposed model achieves strong performance across evaluation metrics such as Dice coefficient, IoU, precision, and recall, indicating reliable segmentation capability. The improvements align with prior research showing that stacking multiple U-Net architectures enhances segmentation quality by refining intermediate predictions and improving generalization .

Overall, this project presents a robust, scalable, and extensible segmentation framework for medical imaging. It addresses key limitations of standard U-Net models and establishes a strong foundation for future advancements, including integration of attention mechanisms, lightweight architectures, and real-time clinical deployment systems.

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